Time series decomposition

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Before you start

This is an in-class lab: work through the material below at your own pace (in pairs or small groups is fine) — I'll walk around to help, and we'll stop along the way to discuss as a group.

This builds on the moving averages, classical decomposition, X11/SEATS and STL material from the decomposition chapter. Today you apply it: decompose real series into trend-cycle, seasonal and remainder components, compare methods, and interpret what you get.

```
library(fpp3)
library(tidyverse)
```

Cheat sheet: decomposition methods

Method	Syntax
Classical	`model(classical_decomposition(variable, type = "additive"))`
X11	`model(X_13ARIMA_SEATS(variable ~ x11()))`
SEATS	`model(X_13ARIMA_SEATS(variable ~ seats()))`
STL	`model(STL(variable ~ trend(window = ...) + season(window = ...), robust = ...))`

After fitting, use `components()` to extract the trend-cycle/seasonal/remainder columns, then `autoplot()` to plot them.

Task 1: Offshore wind power

Use the `windpower` data from `ban430data` (the same one from the Time series graphics lab) again — this time for decomposition.

```
# install.packages("remotes")
remotes::install_github("holleland/ban430data")
library(ban430data)
data(windpower)
```

a) Create a tsibble containing the **daily** wind power data from **Sørlig Nordsjø 2** only (aggregate `windpower` from hourly to daily with the mean, same as in the graphics lab).

b) Decompose the series into trend-cycle T_t, season S_t and remainder R_t, using a suitable decomposition method. Which method(s) from the cheat sheet above are *not* usable here, and why? Why did you choose the method you did?

c) We can also use **STL features** to assess and compare h*ow* seasonal or trending a series is rather than just eyeballing a decomposition (see Section 4.3 of the textbook). Aggregate `windpower` to **monthly** data for **both** locations (like `wp_month` in the graphics lab), then run:

```
windpower %>%
  group_by_key() %>%
  index_by(yearmonth = yearmonth(datetime)) %>%
  summarise(powerprod = mean(powerprod, na.rm = TRUE)) %>%
  features(powerprod, feat_stl)
```

Which location has the stronger `seasonal_strength_year`? What about `trend_strength`?

Task 2: Norwegian wholesale and retail sales index

The data is a monthly index for *Retail trade, except of motor vehicles and motorcycles*, from Statistics Norway (Jan 2000 onwards). It's the `wholesale_raw_lab` object from `ban430data`.

```
library(ban430data)
data(decomposition)

wholesale <- wholesale_raw_lab
head(wholesale, 3)
```

```
# A tibble: 3 x 2
  month   `wholesale and retail sales index`
  <chr>                                <dbl>
1 2000M01                               39.9
```

```
2 2000M02                            38.9
3 2000M03                            42.4
```

a) Convert the `month` column to a `yearmonth` type and turn `wholesale` into a tsibble.
b) Make a time plot.
c) Decompose the series using the classical, X11, SEATS and STL methods. Can you detect any prominent differences between the methods?
d) Try adjusting the trend and season windows of the STL (default values are 21 and 11 respectively). What happens?
e) Using X11 or SEATS - the "statistics agency methods" from the cheat sheet, and what Statistics Norway itself actually uses - plot your seasonally-adjusted time series.
f) Statistics Norway publishes its own official seasonally-adjusted version of this exact series - it's the `wholesale_sa` object in `ban430data` (column `index_sa`; same table, same industry filter, official `VerdiinSesong` content code). Plot it alongside your series from (e). Do they agree?
g) Using your method of choice, plot the detrended series.
h) *Optional*: Implement your own additive classical decomposition by hand (moving averages + seasonal averaging), and compare to `classical_decomposition()`. (See Chapter 3 exercises, exercise 6 if you want to check your approach against a worked solution.)

Wrap-up discussion

For both series today: did the different decomposition methods roughly agree, or did they tell noticeably different stories? When might you prefer STL over classical decomposition, or vice versa? What made X11/SEATS unsuitable for the daily wind power data?